UC8: ROBOTIC ARM CONVEYOR PICK-AND-PLACE WITH OBSTACLE AVOIDANCE

Milestone 1 Official Defense & Technical Evaluation (Week 05)

PROJECT METADATA

- Course: Robotic Project — Vietnamese-German University
- System: UR10e (6-DOF) + 3.6m Conveyor + 2-Finger Claw + 3 Bins
- Evaluation: Milestone 1 Official Defense (Requirements R1 - R4)
- Next Phase: Week 06 Sprint & Milestone 2 (Static Obstacle Avoidance)
- Empirical Status: 100% Verified Closed-Loop Sequencer (70/70 Trials OK)

MILESTONE 1 VERIFIED HIGHLIGHTS

✓ UR10e Motion Streaming: 50 Hz /joint_states driving 6 ArticulationBodies
✓ Industrial 2-Finger Claw: Procedural flange mount, status LED, aluminum arms
✓ Autonomous Sequencer: 7-step closed-loop state machine (zero human input)
✓ Zero-Teleport Physics: Pure friction coasting stop halting cubes (±3mm precision)
✓ Quality Assurance: 70 consecutive trials logged with 100% success rate

Agenda of Milestone 1 Defense & Roadmap

No.	Topic	Core Content & Engineering Focus
01-02	Project Title & Agenda	Context, requirements, and presentation structure
03-04	Baseline Recap & Scope Fulfillment	100% completion of Milestone 1 commitments (R1-R4)
05	System Architecture	Bidirectional closed loop: Unity 2022.3 <-> ROS-TCP <-> ROS Noetic
06	Feedback & Issue Resolutions	Faculty claw feedback, anti-miss scanner, orthogonal zero-sink grasp
07-08	Workcell Layout & Kinematics	Option 1 bilateral symmetry, 90° wrist lock, and Interactive Trajectory [T]
09-10	PhysX Simulation & State Machine	Friction deceleration, multi-item flow, 7-step sequencer
11-12	Benchmark QA & Technical Defenses	70 empirical trials, 100% success rate, answers to faculty committee
13-14	Week 06 Objectives & Milestone 2 Roadmap	R5 conveyor robustness, vision classification, static obstacle avoidance
15	Summary & M1 Acceptance Conclusion	Milestone 1 criteria sign-off and transition to Milestone 2 development

Week 04 Baseline and Commitments Set Out for Milestone 1

Baseline Achieved in Week 04

✓ ROS-Unity bridge: TCP communication established on port 10000 via ros_tcp_endpoint.

✓ Coordinate conversion: Unity coordinates mapped to ROS FLU: X_ROS = Z_Unity, Y_ROS = -X_Unity, Z_ROS = Y_Unity.

✓ Object perception: /detected_object_pose (Pose) and /detected_object_type (String) published at 10 Hz.

✓ MoveIt planning: Pre-grasp feasibility verified in RViz (+15cm hover pose above detected cube base).

Commitments Set Out for Milestone 1

🎯 Trajectory execution: Overcome 'plan-only' barrier; stream real-time joint-states to drive physical UR10e in Unity.

🎯 Grasping mechanism: Implement attach/parenting logic linking target cube to tool0 upon arrival.

🎯 Task controller: Construct autonomous closed-loop sequencer connecting perception, motion, and release.

🎯 Multi-bin sorting: Route red, yellow, and blue cubes to dedicated destination bins without collision.

🎯 Milestone 1 Context: Fulfilling 100% of these commitments in Week 04 is the prerequisite for Milestone 1.

Milestone 1 Scope: 100% Commitment Fulfillment Status

100% Commitment Fulfillment Status

✓ Trajectory Execution (R2/R3): Streamed /joint_states at 50 Hz to 6 ArticulationBodies with zero sagging and smooth tracking (100% DONE).

✓ Grasping Mechanism (R4): Implemented industrial Attach/Parenting logic in GripperController.cs with instant latching (100% DONE).

✓ Closed-Loop Sequencer (R4): Constructed 7-step autonomous sequencer (task_state_machine.py) with zero human intervention (100% DONE).

✓ 3-Bin Sorting System (R4): Deployed Red (Left), Yellow (Right), and Blue (Rear) bins with dedicated non-singular trajectories (100% DONE).

Milestone 1 High-Value Enhancements

⚙ Kinematic Optimization: Locked wrist strictly downward (q4 = 90.0°) & implemented pure base SCARA sweep, cutting cycle time by 65%.

⚙ Interactive Trajectory Trail [T]: Implemented real-time neon ribbon visualizer at tool0 with 3.5s auto-fade and color-coded destination paths.

⚙ Realistic PhysX Deceleration: Replaced artificial freezing with pure friction coasting (StopConveyor()) halting cubes at X ≈ 0.00 m (± 3 mm).

⚙ Continuous Multi-Item Flow: Upgraded Spawner to maintain 2-3 items on conveyor simultaneously with safe 38cm spacing.

⚙ Empirical QA Logging: Integrated automated CSV logger recording 70 consecutive trials with 100% reliability for Milestone 1 defense.

System Architecture: Bidirectional Communication Pipeline

Layer	Key Responsibility	Topic / Protocol	Rate & Direction
Unity Perception	Detects cube, halts conveyor, converts frame to ROS FLU	/detected_object_pose, /detected_object_type	10 Hz (Unity -> ROS)
ROS-TCP-Endpoint	Low-latency network bridge brokering serialised ROS messages	TCP Port 10000 Socket Streaming	Real-time (Bidirectional)
State Machine	Closed-loop task logic sequencing 7 execution states	task_state_machine.py orchestrator	Event-driven (ROS Node)
MoveIt Core	Plans collision-free Cartesian paths & joint trajectories	manipulator MoveGroupCommander	MoveIt <-> State Machine
Robot Controller	Maps joint positions to 6 UR10e ArticulationBodies	/joint_states (JointTrajectoryPoint)	50 Hz (ROS -> Unity)
Gripper Controller	Parenting attach logic; animates fingers and clamps cube	/ur10e/gripper_command, /gripper_status	Event-driven (ROS <-> Unity)

CORE INTERFACE SPECIFICATIONS & STANDARDS

- Coordinate Mapping: X_ROS = Z_Unity, Y_ROS = -X_Unity, Z_ROS = Y_Unity
- Planning Group: manipulator | Planning Frame: base_link | End Effector: tool0
- Streaming Topics: /detected_object_pose (Pose), /detected_object_type (String), /joint_states @ 50 Hz, /ur10e/gripper_command (Bool)

Instructor Feedback and Solving Runtime Engineering Challenges

Instructor Feedback

Requirement: Visible 2-Finger Claw

• Faculty Feedback: Instructor requested a visible mechanical gripper claw on the robot rather than an abstract suction/contact point.

• Procedural Claw (VisualClaw): Designed flange adapter (1.2cm), navy housing chassis (2.0cm), and dual-color status LED (Cyan/Emerald).

• Industrial Fingers: Aircraft aluminum arms (4.0cm) with high-friction black rubber pads; animated carriage (9.2cm to 6.1cm span).

Runtime Issue 1

Challenge: Missed Cubes in Continuous Flow

• Observed Bug: Over extended runs, cubes occasionally slipped past pick station due to missed trigger collisions.

• Root Cause: OnTriggerEnter is fragile under continuous multi-item flow and physics frame discretization.

• Technical Solution: Replaced trigger collisions with continuous active scanning (FindLeadingCubeOnConveyor()) + safety stop latch at X >= -0.015m (100% reliable).

Runtime Issue 2

Challenge: Conveyor Penetration & Off-Axis Twist

• Observed Bug: Claw slightly penetrated belt during descent; upon grasping, cube twisted off-axis by 13.6°.

• Root Cause: Tool orientation was skewed (0.7854 quaternion), and forced position/rotation snapped cube off-axis.

• Technical Solution: Standardized down_orientation = Quaternion(0.7071, 0.7071, 0, 0), elevated fingertips (+2.3cm clearance), and pure parenting.

Workcell Layout: Front Pick Zone and Flanking Bins

Component	Selected Setup	Engineering Rationale
Conveyor	3.6 m, centered directly in front (X = -0.60 m)	Stable transport through robot workspace; ample gliding run
Pick zone	Front center (X = -0.60 m, Y = 0.00 m)	Symmetric approach minimizing arm travel distance & inertial moments
Red and yellow bins	Left (Y = +0.55 m) and right (Y = -0.55 m)	Clear transfer paths eliminating collision hazards with conveyor
Blue bin	Behind robot (X = +0.50 m, Y = 0.00 m)	Separates third sorting target via pure base shoulder pan rotation
Support surface	4.0 m × 2.0 m heavy industrial table	Rigid mechanical foundation for rails, bins, and robot fixtures

SPATIAL CLEARANCE & SAFETY DIMENSIONS

- Bin Mouth Enlargement: Expanded +35% to 0.38 m × 0.38 m × 0.10 m for 100% free-fall capture rate.
- Vertical Safety Envelopes: +16 cm Pre-grasp & Lift altitude completely clears conveyor aluminum rails (Z = +0.10 m) with zero contact.

Downward Wrist Lock, Transfer Motion & Trajectory Trail

Motion Strategy

✓ Downward tool orientation: tool0 remains vertical ([0, 0, -1]) during approach, grasp and release via locked wrist (q4 = 90.0°).

✓ Pre-grasp waypoint: The robot approaches above the target (+16cm) before descending.

✓ Vertical descent: The final approach moves along pure Cartesian Z, avoiding contact with conveyor rails.

✓ Lift before transfer: The cube clears the conveyor (+16cm) before base rotation begins.

✓ Dominant shoulder pan: The transfer uses a large base rotation while other joints remain stable like a SCARA crane.

Joint Values across Poses (Radians)

⚙ Standby pose: [0.2945, -1.4760, -2.1278, -1.1022, 1.5708, 0.5291]

⚙ Red bin (left): [-1.0741, -1.4731, -2.2727, -0.9602, 1.5708, 0.5291]

⚙ Yellow bin (right): [1.7077, -1.4731, -2.2727, -0.9602, 1.5708, 0.5291]

⚙ Blue bin (rear): [-2.7858, -1.3576, -2.3651, -0.9833, 1.5708, 0.5291]

⚙ Kinematic Invariance: Shoulder lift, elbow, and wrist joints stay within 0.15 rad across all bins; only shoulder pan swings widely.

INTERACTIVE TRAJECTORY VISUALIZATION & ANTI-CLUTTER TRAIL (HOTKEY [T])

- Interactive Toggle [T]: Press [T] during runtime to toggle smooth 3D neon ribbon trail mapped to tool0 end-effector.
- Anti-Clutter Auto-Fade (3.5s): Automatically fades out tail points after 3.5s, preventing visual clutter across long runs.
- Color-Coded Target Paths: Dynamically tints trail based on destination bin: Red (#FF2A6D), Yellow (#FFD166), Blue (#05D9E8).

Conveyor Motion and Pick-Zone Stopping

Runtime Behaviour & Friction Deceleration

✓ Transport: Objects move at configured conveyor speed (rock-steady 0.30 m/s).

✓ Detection: Pick-zone sensor actively identifies leading SortableObject cube.

✓ Stop: At X >= -0.015 m, conveyor cuts power. Friction halts cube smoothly at X ≈ 0.000 m (± 3 mm).

✓ Zero Teleportation: Strictly eliminates artificial coordinate snapping or kinematic freezing.

✓ Resume: Conveyor restarts automatically after cube handling and release into bin.

Multi-Item Flow & Visual Rig Consistency

⚙ Continuous Multi-Item Flow: Upgraded Spawner maintains 2-3 active cubes on belt simultaneously, providing continuous sorting.

⚙ Anti-Stacking Spacing: Strict 38 cm minimum spacing (IsSpawnAreaClear()) prevents any cube collisions at spawn head.

⚙ Visual consistency: Belt texture scrolling is synchronized 1:1 with physical object motion at 0.30 m/s.

⚙ Mechanical Rig Structure: Ribbed rubber belt, dual aluminum guide rails, dual chrome rollers, and gearmotor unit.

⚙ Optical Sensor LED: Dual-color indicator displays Green when running, switching to Red when cube stops at pick center.

7-Step Closed-Loop Task State Machine Execution

Step	State Name	Primary Action	Safety & Validation Guard
1	STANDBY	Arm hovers above pick station (Z = 0.42 m)	Elbow held elevated; wrist strictly downward
2	APPROACH	Rapid Cartesian descent to Pre-grasp Pose (+16cm)	Clears conveyor rails in < 0.5s; wait=True feedback
3	DESCENT	Vertical Cartesian descent to contact (+6.5cm)	Fingertips float 2.3cm above belt (zero sinking)
4	GRASP	Publish gripper command (True); latch object	Attach logic latches cube; MoveIt updates PlanningScene
5	LIFT	Linear Cartesian lift back to safe height (+16cm)	10cm vertical lift clears conveyor rails without snagging
6	MOVE_TO_BIN	Single shoulder pan rotation to destination bin	Non-singular trajectory avoids table collisions
7	RELEASE	Publish gripper command (False); release cube	Parenting detached; gravity drops cube into bin
8	READY / REARM	Arm returns to Standby pose over pick zone	Resumes conveyor & triggers spawner for next cycle

Empirical Test Benchmark: 70 Consecutive Trials

Performance Metric	Requirement	Empirical Measured Result	Status Evaluation
Total benchmark trials	>= 10 cycles	70 consecutive cycles logged	Exceeds requirement by 700%
Grasp success rate	>= 90.0%	100.0% (70 / 70 trials)	Perfect reliability; zero drops
Bin sorting accuracy	>= 90.0%	100.0% (70 / 70 trials)	100% correct bin routing
Mean cycle time	< 30.0 s	11.23 s (Min: 10.8s, Max: 12.1s)	2.7x faster than threshold
Conveyor transport speed	0.2 - 0.5 m/s	0.30 m/s rock-steady velocity	Optimal industrial pace
Coasting stop deviation	<= 20 mm	+- 3 mm repeatable halt	High physical precision

Faculty Committee Engineering Defenses

Why are Pre-grasp and Lift poses necessary?

• Industrial workcells divide workspace into Free-space transit and Fine Cartesian approach.

• Pre-grasp (+16cm) establishes a collision-free holding waypoint above the moving conveyor.

• Moving directly on diagonal paths risks collision with conveyor rails or bin edges.

Why avoid fixed sleep delays for motion timing?

• Motion time varies with distance (1.07 rad to Red bin vs 2.78 rad to Blue bin).

• Fixed sleep delays cause premature grasp triggers (dropped objects) or unnecessary idle time.

• MoveIt trajectory execution feedback (wait=True) confirms physical arrival before next step.

How were ragdoll sagging & oscillations eliminated?

• 1. base_link locked with immovable = true in ArticulationBody.

• 2. Gravity disabled (useGravity = false) on arm links; gravity compensation handled by MoveIt dynamics.

• 3. Tuned high PhysX drive stiffness (stiffness = 50000, damping = 1000, forceLimit = 10000).

Week 06 Sprint: Fulfilling Requirement R5 & Robustness

R5 Conveyor Robustness

• 5-Minute Continuous Operation: Validate that conveyor, spawner, and sensor run non-stop without jamming.

• Variable Velocity Modulation: Support dynamic speed adjustment (0.1 to 0.5 m/s) via ROS parameter service.

• PhysX Friction Stability: Eliminate micro-slip across diverse cube weights (0.1kg - 0.5kg).

Deterministic Spawner

• PRNG Seed Reproducibility: Implement seed parameters for RandomObjectSpawner.cs.

• Guaranteed Test Batches: Allow exact sequence replay (e.g. Red -> Blue -> Yellow) for automated regression.

• Spacing Latch Verification: Guarantee 38cm spacing under all belt acceleration profiles.

Fault Recovery

• Belt Jam Auto-Recovery: Automated reverse-and-clear logic if an unexpected stall occurs.

• Perception Timeout Guard: Fail-safe state machine rollback if object detection frame drops.

• Socket Watchdog: Auto-reconnect handling for ROS-TCP-Endpoint in case of network hiccups.

WEEK 06 DEFINITION OF DONE

• Complete fulfillment of Requirement R5 (Conveyor & Spawner Integration).
• The workcell runs hands-free for > 5 minutes with zero cube miss, zero jam, and deterministic regression logging.

Milestone 2 (Week 10): Static Obstacle Avoidance & Perception

R5 & R6: Conveyor & Vision

• R5 (Continuous Conveyor): Industrial belt running continuous multi-item queues with zero jam.

• R6 (Computer Vision Pipeline): Camera perception module replacing metadata detection.

• Color Classifier: OpenCV HSV segmentation classifying Red, Blue, and Yellow cubes under varying lighting.

R7: 3D Localization

• Camera-to-Robot Calibration: Hand-eye calibration linking camera pixel coordinates to UR10e base_link.

• Centroid Extraction: Projecting bounding box centroids to 3D Cartesian coordinates.

• Localization Accuracy: Benchmarked measurement error <= 20mm compared to ground truth.

R8: Static Obstacle Avoidance

• Static Wall Barrier: Introducing physical obstacle wall between pick station and sorting bins.

• MoveIt PlanningScene: Real-time obstacle representation in collision matrix.

• Collision-Free Detour: OMPL planner automatically computes curved bypass paths around the wall.

MILESTONE 2 (WEEK 10) ACCEPTANCE CRITERIA: STOP-AND-PICK MODE

• Verification of Requirements R5, R6, R7, and R8 in Stop-and-Pick mode (conveyor stops before grasp).
• Arm successfully navigates around static obstacle barrier with zero collision across >= 20 benchmark trials.

Summary of Deliverables & Official Acceptance Sign-Off

Requirements Met

✓ R1 (Architecture & Scope): 100% Approved

✓ R2 (UR10e MoveIt Planning): 100% Operational

✓ R3 (ROS-Unity Bridge): 100% Low-Latency Streaming

✓ R4 (Closed-Loop Pick & Place): 100% Autonomous

Empirical Evidence

✓ 70 Consecutive Trials Logged

✓ 100.0% Grasp Success Rate

✓ 100.0% Multi-Bin Placement Rate

✓ 11.23s Mean Cycle Time (2.7x faster)

Deliverables Package

✓ Git Tag Release: m1-verified

✓ Interactive Trajectory Trail [T]

✓ Complete trials.csv QA Evidence

✓ Multi-Camera Demo Visuals

FORMAL MILESTONE 1 ACCEPTANCE REQUEST

- All commitments made for Milestone 1 have been 100% fulfilled and reinforced with faculty feedback solutions.
- The UC8 sorting workcell is fully verified and ready for formal Milestone 1 sign-off and progression to Week 06 / Milestone 2.